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API Remoting for `llama.cpp`: Near-Native GPU Speed in `macOS` Containers

Kevin Pouget, Jose Castillo Lema – January 31st, 2026
Red Hat Principal Software Engineer

FOSDEM - AI Plumbers track

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Containers in macOS ???

But

, right ?

And

, no ???

So we need a **virtual machine** to bring containers to macOS

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MacOS virtual container stack

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libkrun

Libkrun is a tiny, embeddable VM runtime library that uses the host's native hypervisor APIs:

- macOS → Apple Hypervisor / Virtualization.framework
- Linux → KVM

MacOS virtual container stack

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Podman AI Lab is the easiest way to work with LLMs on your local developer workstation.
Find a catalog of recipes, leverage a curated list of open source models, experiment and compare the model.

ramalama

Make working with AI boring through the use of OCI containers

```
ramalama pull llama3.2
ramalama run  llama3.2
ramalama bench llama3.2
```

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what about GPU acceleration ?

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what about GPU acceleration ?

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- Linux container GPU acceleration is straightforward
  - just expose the required `/dev` files and userland libraries
- Virtual GPU acceleration **is fundamentally different**
  - **Full device passthrough?** not possible for the main display
  - **Hardware-assisted device passthrough?** not possible without … hardware assistance
  - **Virtual device emulation?** Not efficient for GPU acceleration
  - **Paravirtualized device?** Let's talk more about this!

    (current solution for Vulkan acceleration & the API Remoting approach)

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MacOS Virtual GPU stack

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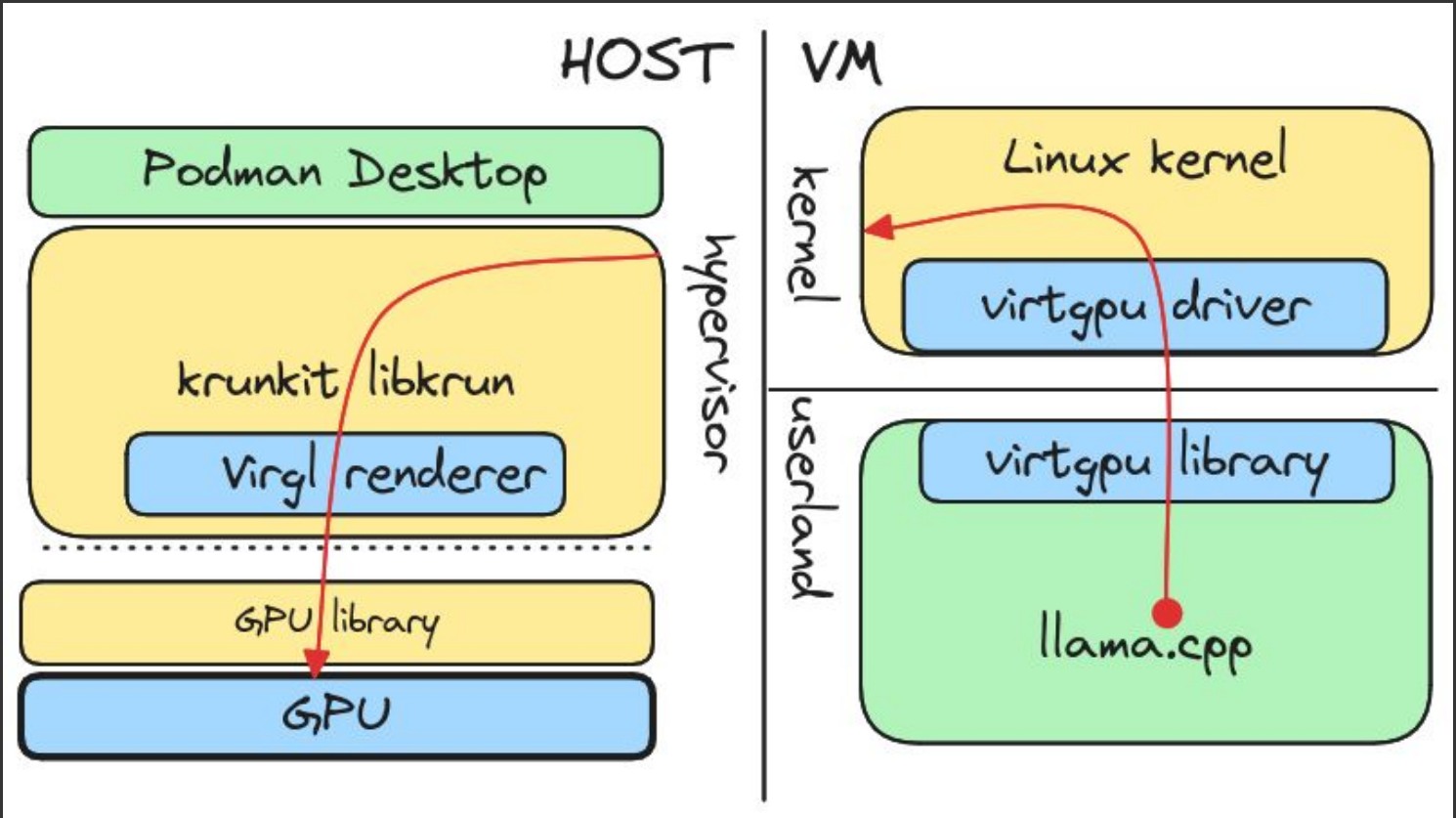

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MacOS Virtual GPU stack

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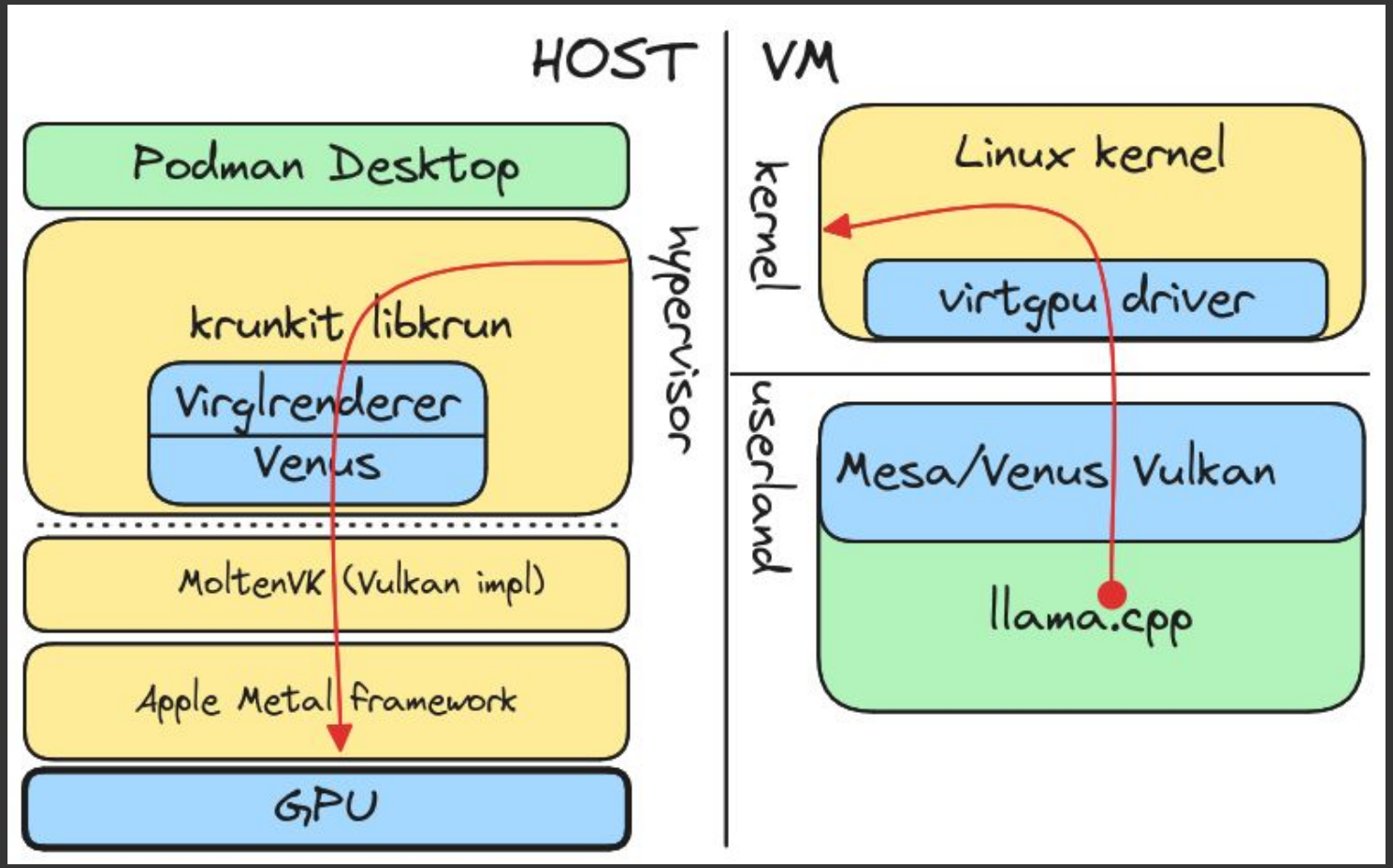

MacOS Virtual GPU stack performance

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MacOS Virtual GPU stack performance

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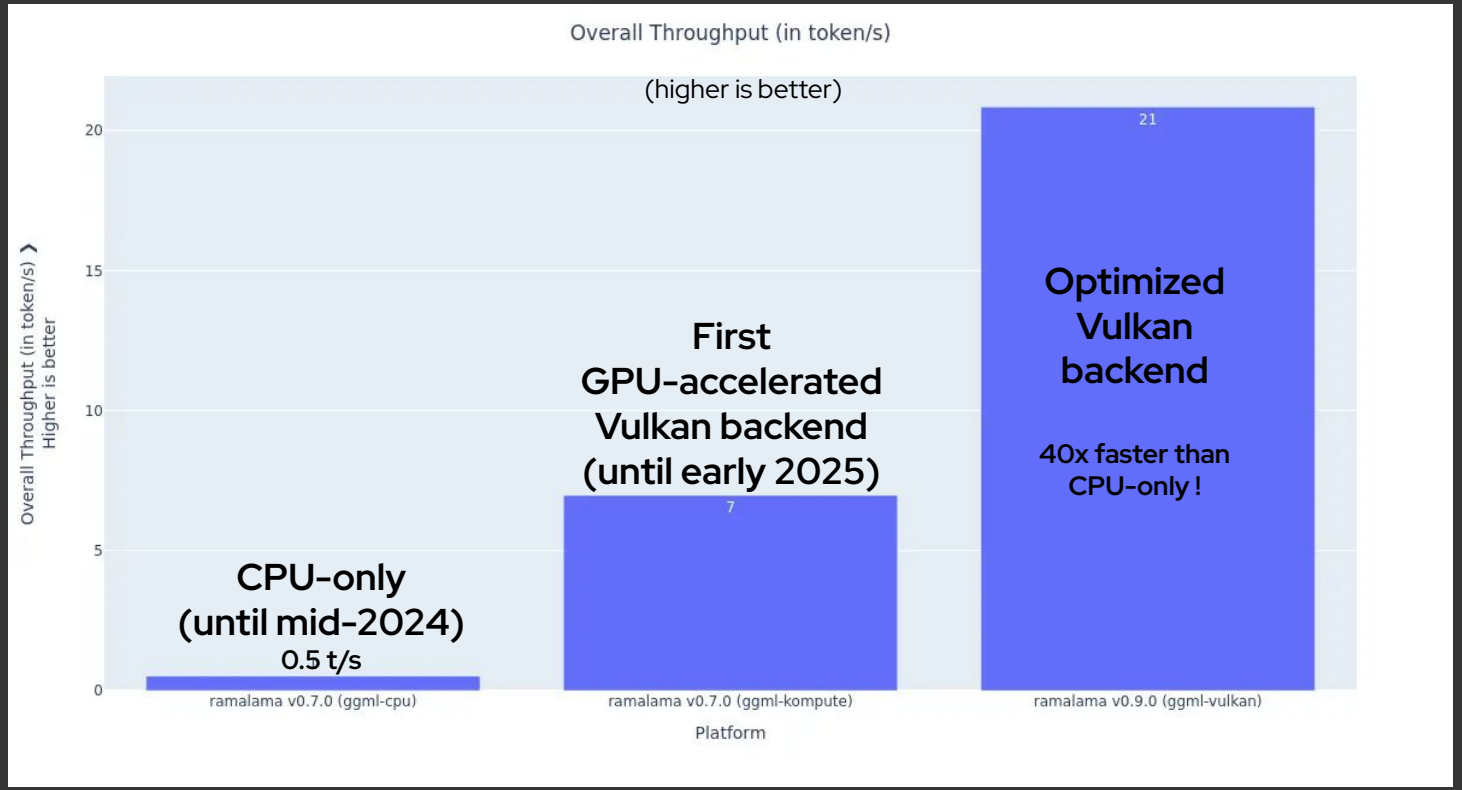

How we improved AI inference on macOS Podman containers

(developers.redhat.com)

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MacOS Virtual GPU stack performance

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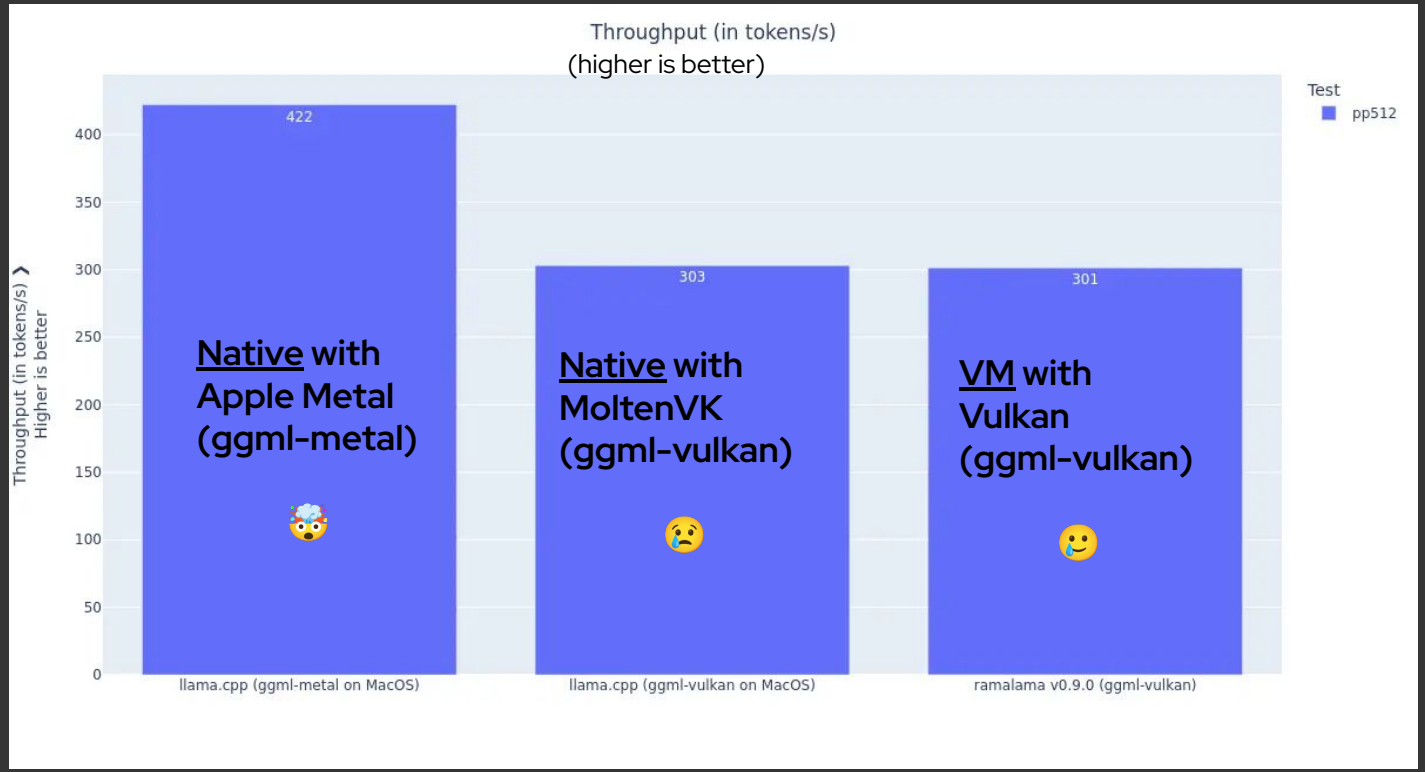

- Vulkan *MoltenVK* is the bottleneck :/
- New Mesa KosmicKrisp for MacOS Vulkan

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How to improve the MacOS Virtual GPU stack performance ?

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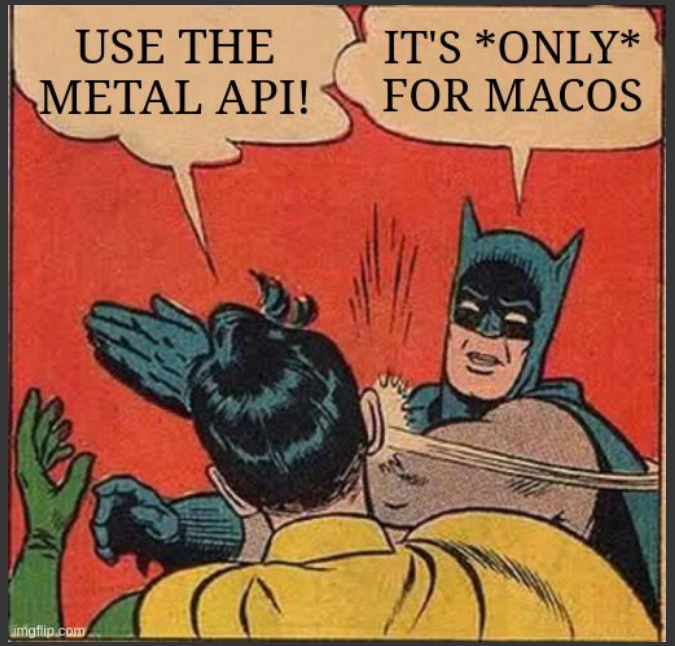
How to improve the macOS virtual GPU stack performance ?



- Deeply specific to MacOS
- Apple Swift
- legal aspects? 🤷‍♀️

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How to improve the macOS virtual GPU stack performance ?

- GGML is the tensor library backend interface for `llama.cpp`
- Use GGML-Metal on MacOS
- Trigger the GGML API calls from the guest, execute them on the host

  ⇒ API Remoting!

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API Remoting

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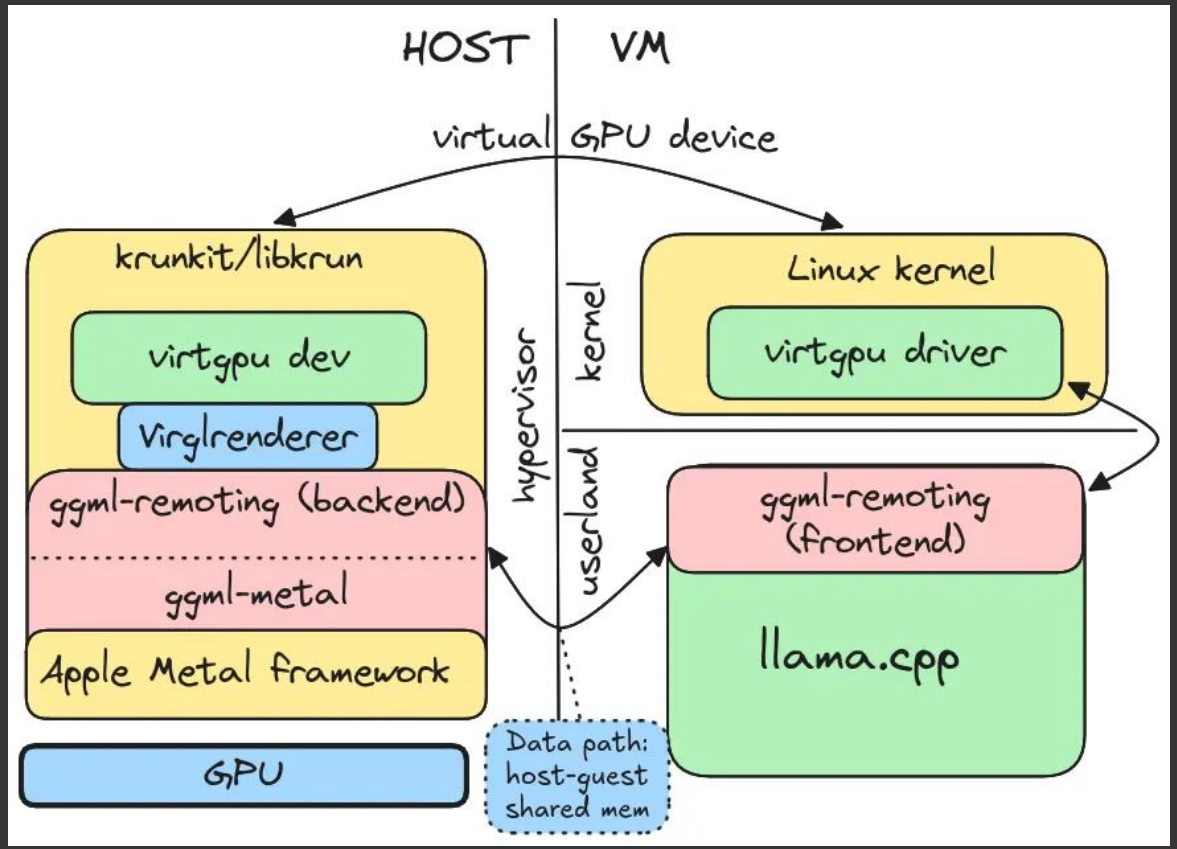

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API Remoting

- Intercept the API calls in the guest
- Forward the parameters to the host
- Call the actual API in the host
- Return the data back to the guest

- 👎 Not a generic solution, specific to llama.cpp use case
- 👎 Pay the price of the hyper-calls latency + synchronization
- 👍 Use para-virtualization (host-guest shared memory pages) to avoid copying the data buffers

- Technique already used for the Vulkan-Venus communication

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API Remoting POC

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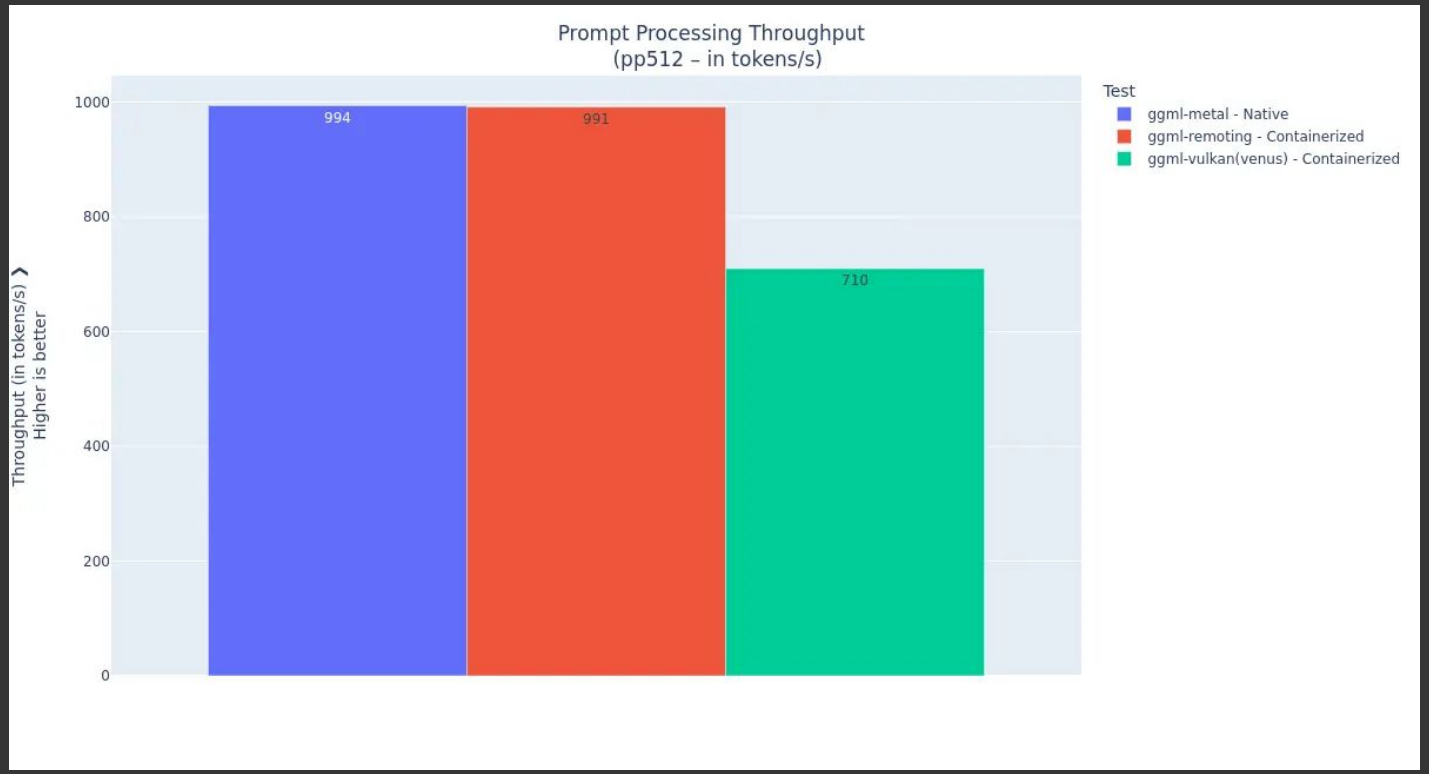

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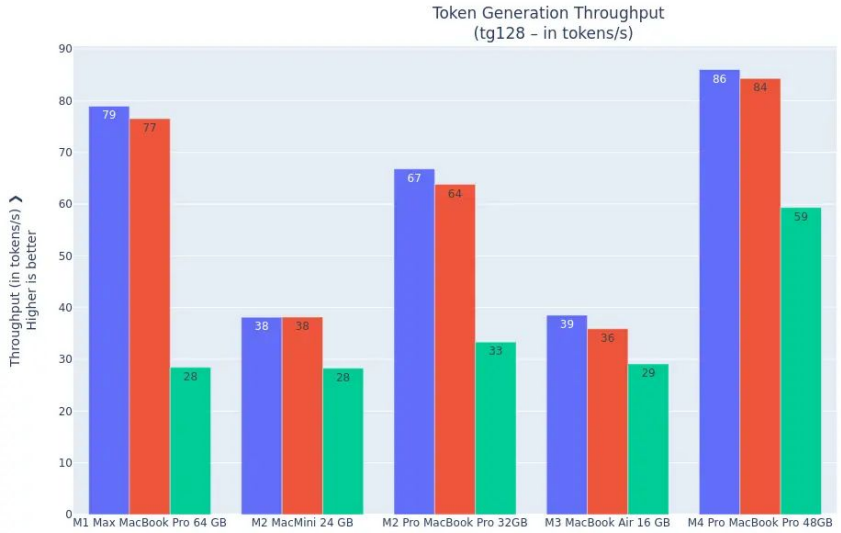
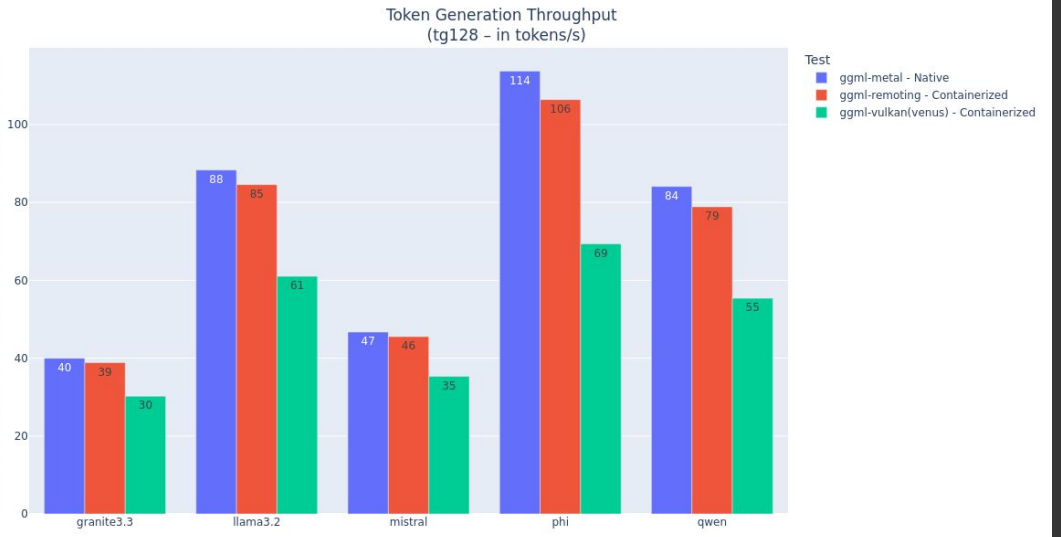
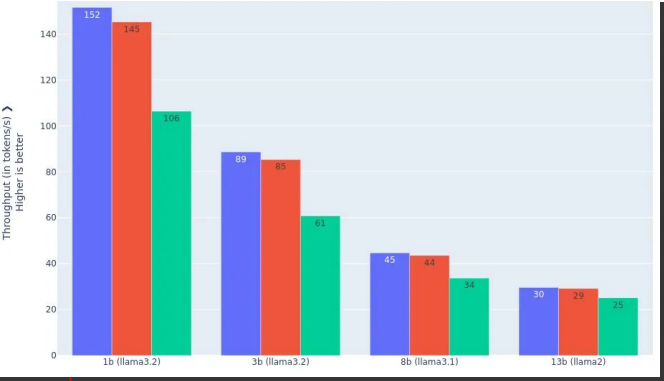
API Remoting POC







- Tested with different hardware
- Tested with different model families
- Tested with different model sizes

  ⇒ smaller models have more overhead (expected)

  - 151.59 t/s ⇔ 1 token every 6.6ms
  - 145.29 t/s ⇔ 1 token every 6.7ms

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API Remoting Upstreaming status

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`llama.cpp` (GGML frontend and backend)

- ggml: new backend for Virglrenderer API Remoting acceleration
  ⇒ merged earlier this week (Jan 26th) 🎉

`virglrenderer` (generic APIR component)

- Introduce an API Remoting trampoline component
  ⇒ review pending

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API Remoting

- Reach native speed with MacOS llama.cpp container inference (developers.redhat.com)
  - Try with Podman Desktop ⇒ built an extension for easy testing
  - Try with Ramalama
    - Benchmark with Ramalama
  - Try it on Linux
    - with `krun` OCI container runtime
  - Try it on Windows
    - ~~kidding~~ 😛
    - well, not ready yet, still an early prototype

  - Integration in Podman Desktop
    - expected in the next quarter (Q2-2026)

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API Remoting Security considerations

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Key advantages

- the hypervisor (`krunkit`) runs with simple user privileges
- the backend and the GPU only execute trusted code
  - This eliminates risks coming from the execution of malicious kernels

Current limitations and considerations:

- The backend library currently lives within the hypervisor address space
  - a crash within the backend library could bring down the hypervisor.
  - the backends of multiple containers run in the same process (usual per-process GPU isolation not available)

Overall, we can say that this API remoting design isn't **multi-tenant safe.**

llama.cpp / Ramalama nightly CI

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entry_id	platform	containerized	version	pp_throughput	tg_throughput	llm_load_test_ttft	llm_load_test_itl
3	macos/llama_cpp/metal	False	b7770	415 tokens/s	39 token/s	1147 ms	25 ms
2	macos/llama_cpp/vulkan	False	b7770	241 tokens/s	28 token/s	2732 ms	46 ms
1	podman/ramalama	True	main-260115.1614	220 tokens/s	27 token/s	2684 ms	159 ms

Performed 12 KPI regression analyses over 3 entries x 4 KPIs. 0 KPIs didn't pass.

(link)

- Misc. bug: performance regression in llama-server (ggml-vulkan) #17033
- Eval bug: Vulkan not working on Intel GPU #17056

entry_id	platform	containerized	version	pp_throughput	tg_throughput	llm_load_test_ttft	llm_load_test_itl
4	macos/llama_cpp/metal	False	b6933	415 tokens/s	40 token/s	1133 ms	25 ms
3	macos/llama_cpp/vulkan	False	b6933	223 tokens/s	28 token/s	7459 ms	62 ms
1	podman/ramalama	True	main-251101.0607	223 tokens/s	28 token/s	3815 ms	42 ms
2	podman/ramalama/no-gpu	True	main-251101.0607	24 tokens/s	19 token/s	19192 ms	57 ms

Performed 16 KPI regression analyses over 4 entries x 4 KPIs. 5 KPIs didn't pass.

(link)

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API Remoting

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Control path (guest side)

```
void ggml_device_get_memory(ggml_backend_dev_t dev, size_t * free, size_t * total) {

  // the encoder and decoder are shared memory buffers
  REMOTE_CALL_PREPARE(gpu, encoder, APIR_CMD_DEVICE_GET_MEMORY);

  apir_encode_ggml_dev(encoder, dev);

  REMOTE_CALL(gpu, encoder, decoder); // blocking call triggering the remote execution

  apir_decode_size_t(decoder, &free);
  apir_decode_size_t(decoder, &total);
}
```

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Data path initialization (buffer_type_alloc_buffer**)**

```
// guest side
virtgpu_shmem_create(gpu, size, &ctx.shmem)
apir_encode_virtgpu_shmem_res_id(encoder, ctx.shmem.res_id);
apir_encode_size_t(encoder, &size);

// host side
apir_decode_virtgpu_shmem_res_id(dec, &shmem_res_id);
shmem_ptr = ctx->iface->get_shmem_ptr(ctx->ctx_id, shmem_res_id); // to the hypervisor
// to the actual llama.cpp GGML backend
buffer = dev->iface.buffer_from_host_ptr(dev, shmem_ptr, size, max_tensor_size);
```

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Data path transfers (guest side only)

```
// set_tensor
memcpy(shmem_ptr + offset, data, size);

// get_tensor
memcpy(data, shmem_ptr + offset, size);`
```

API Remoting

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  - Try with Ramalama:

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```
# download and extract the tarball and enter its directory
bash ./update_krunkit.sh # update and re-sign the binaries
# restart the VM with the custom virglrenderer library
bash ./podman_start_machine.api_remoting.sh

export CONTAINERS_MACHINE_PROVIDER=libkrun
export IMG=quay.io/crcont/remoting:v0.16.0-apir…
ramalama run --image $IMG llama3.2
```

https://github.com/crc-org/llama.cpp/releases

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  - Benchmark with Ramalama
    ⇒ reproduce the benchmark figures yourself!

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```
ramalama bench --image $IMG llama3.2  # API Remoting performance
ramalama --nocontainer bench llama3.2 # native Metal performance
ramalama bench llama3.2               # Venus/Vulkan performance
```

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```
ramalama run --oci-runtime krun --image $IMG llama3.2
```

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    - kidding 😛

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